Two Month Internship Report

On

Data Analysis using Python and Power BI

Submitted By:

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DECLARATION

I, **Ms Gunjan Ajay Pandya** hereby declare that the Internship report entitled

"Data Analysis using Python and Power BI" is an original work conducted and prepared by me. This document is submitted in partial fulfillment of the requirements for the Internship Program at Indian Institute of Information Technology. I affirm that this report is a result of my efforts and contributions. Any reference to exist in research, direct quotations, or paraphrasing has been properly acknowledged. I understand the importance of this declaration and hereby certify that the information presented in thisreport is true and accurate to the best of my knowledge and belief.

Date :31/7/25 Gunjan Ajay Pandya

CERTIFICATE

This is to certify that the **Gunjan Ajay Pandya** student of **Ramdeobaba College of Engineering and Management** has completed her Two month Internship and submitted Internship Report on the topic: "Data Analysis using Python and Power BI"under my supervision during 1/6/25 –31/7/25 in the Department of Computer Science& Engineering Indian Institute of Information Technology, Nagpur

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ACKNOWLEDGEMENT

The two Internship opportunity I had with Indian Institute of InformationTechnology,Nagpur was a great chance for learning and professional development. I would like to express my deepest thanks and great sense gratitude towards Dr.Santosh Sahu sir for his valuable guidance, timely help and constant encouragement during my internship of two months on the topic Data analysis using python and Power BI . I extend my heartfelt thanks to all members of IIIT, Nagpur for providing me resources essentials for carrying out this internship. I perceive this opportunity as a big milestone in my career development.I will strive to use gained professional skills and knowledge in the best possible way, and I will continue to work on their improvement, in order to attain desired career objectives.

Gunjan Ajay Pandya

ABSTRACT

During my internship at the Indian Institute of Information Technology, Nagpur (IIIT Nagpur), I worked on a range of data analytics projects across diverse domains such as economic analysis, electric vehicle market evaluation, and e-commerce performance tracking. My primary focus was on leveraging tools like Microsoft Power BI and Python to clean, analyze, and visualize large datasets through interactive dashboards.

In the GDP analysis project, I used Python to preprocess and structure country-level economic data and answered some questions based on it. In the Electric Vehicle (EV) sales project, I focused on analyzing state-wise EV adoption across India, comparing the growth of 2-wheeler and 4-wheeler segments. Key metrics such as penetration rates, sales rankings, and year-wise adoption patterns were visualized to assist in understanding market dynamics. The e-commerce dashboard project involved analyzing sales data to understand product performance, customer behavior, and regional revenue trends. By integrating DAX measures, slicers, and drill-down features in Power BI, I created a dynamic dashboard that provided actionable insights for business decision-making.

Throughout these projects, I applied data cleaning, visualization, and analysis techniques to extract meaningful patterns and insights from raw datasets. This internship strengthened my understanding of practical data analytics workflows and improved my ability to turn data into informed, visual decision-support tools.

Keywords: Data Analytics, Python, Power BI, GDP Comparison, EV Sales Analysis, E-commerce Dashboard, Data Visualization, Business Intelligence

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LEARNING OBJECTIVES

1. **Practical Implementation of Data Analytics Concepts**
   The primary objective was to bridge the gap between theoretical learning and practical application. By working on real-world datasets—such as electric vehicle sales, e-commerce , and national GDP trends—I aimed to develop the ability to derive meaningful insights using Python and Microsoft Power BI.
2. **Proficiency in Industry-Standard Tools**
   A key goal was to enhance technical proficiency in data analysis tools. The internship provided an opportunity to work extensively with Power BI for building interactive dashboards, and Python libraries such as pandas, numpy, matplotlib, and seaborn for data preprocessing, analysis, and visualization.
3. **Domain Awareness Across Multiple Sectors**
   Exposure to data from diverse domains—including the automotive industry, retail sector, and economic indicators—allowed me to understand how data analytics supports decision-making across different fields. This objective focused on developing domain-specific awareness and adaptability.
4. **Development of Analytical and Problem-Solving Skills**
   A significant objective was to cultivate a structured and logical approach to analyzing data. Throughout the internship, I engaged in tasks involving data cleaning, trend identification, and insight generation, thereby enhancing my critical thinking and analytical capabilities.
5. **Effective Data Communication and Visualization**
   By designing clear and interactive dashboards, I learned to present complex data in a user-friendly format that supports data-driven decisions for a range of stakeholders.

1. INTRODUCTION

During my internship at the Indian Institute of Information Technology, Nagpur (IIIT Nagpur), I engaged deeply with data science, focusing on data analytics and visualization using both Microsoft Power BI and Python. This experience provided a comprehensive understanding of how data can be leveraged to solve complex business problems and deliver actionable insights.

Over the past few years, we've seen how much the world relies on data—from government policy decisions to what product appears next in your online shopping cart. Data is everywhere, and learning how to use it properly can make a real difference. That's exactly what this internship taught me. Over the course of two months, I got hands-on experience with using Python and Power BI—two powerful tools that help turn raw data into meaningful insights.

I worked on three projects during this time:

1. GDP Analysis
2. Electric Vehicle (EV) Sales Dashboard
3. E-Commerce Performance Dashboard

Each project helped me build and apply my knowledge in a new context—whether it was analyzing the economy, tracking green vehicle adoption, or understanding online shopping trends.

1.1 Description of the Problem Statement

1. **GDP Analysis**

   Understanding a country's economic health is fundamental for policy-makers, investors, and the general public. Gross Domestic Product (GDP) is a critical indicator . However, raw GDP figures across years or countries offer limited value without contextual visual interpretation. The project aimed to answer questions like:

   - How has GDP grown or declined over the years?
   - What is the effect of inflation, foreign investment, or government policies on GDP?
   - How do different countries compare in terms of GDP trends?

2. **Electric Vehicle Sales Analysis**

   With the global push toward sustainable mobility, India has also experienced a surge in electric vehicle (EV) adoption.This project explored:

   - Which states are leading in EV sales?
   - How is the EV penetration rate evolving annually?
   - What seasonal or quarterly trends are observed in sales?

3. **E-commerce Dashboard**

   The rapid growth of online commerce has resulted in vast amounts of customer, sales, and regional data. Businesses need to understand product performance, customer preferences, and regional demand shifts. The dashboard focused on:

   - Top-selling categories and regions
   - Customer behavior and purchase trends
   - Performance comparisons using dynamic filters and KPIs

1.2 Scope of Work

The internship and the corresponding projects involved the complete data analysis lifecycle:

- Data Collection & Import

  Datasets were collected in CSV or Excel formats and were imported into Python and Power BI environments.
- Data Cleaning & Preprocessing

  Using Pandas, I handled missing values, outliers, and inconsistent data. Pivoting, filtering, and transformation were used to create meaningful structures.
- Exploratory Data Analysis (EDA)

  With Matplotlib and Seaborn, I visualized patterns, distributions, and correlations between variables. This helped identify trends and generate hypotheses.
- Dashboard Creation in Power BI

  Data was loaded into Power BI where relationships were built, DAX formulas were applied, and multiple charts, slicers, and drill-downs were integrated to build user-friendly dashboards.
- Insight Generation

  Key questions were answered through visual and statistical insights. Each dashboard was built with usability and interactivity in mind, aimed at business and policy decision-making.

1.3 Methodologies Used for solving the problem

- Data Collection & Integration: Collected datasets from various sources and combined them into a unified structure for analysis.
- Data Preprocessing: Cleaned and transformed data by handling missing values, outliers, and inconsistencies.
- Exploratory Data Analysis (EDA): Analyzed data patterns and anomalies to guide deeper analysis and ensure accuracy.
- Statistical Analysis: Used techniques like regression, clustering, and descriptive statistics to uncover trends and relationships.

- Data Visualization: Built interactive dashboards using Power BI and Python to communicate insights effectively.
- Predictive Analytics: Inferred possible future trends using historical data to support strategic decisions.

1.4 Contribution Proposed by Student

During the internship, I contributed in the following ways:

- Designed Interactive Dashboards to provide real-time insights across different domains.
- Improved Data Visualization Skills by presenting complex data in a clear and simple way.
- Generated Actionable Insights that supported better decision-making and strategy.
- Engaged in Collaborative Learning by working with mentors, sharing ideas, and solving challenges together.

CHAPTER 2
LITERATURE SURVEY

The use of data analytics platforms like Python and Microsoft Power BI has become central to transforming raw datasets into meaningful, interactive insights. During my internship at the Indian Institute of Information Technology, Nagpur, I applied these tools to real-world datasets across domains such as economics, mobility, and retail. This literature survey summarizes key research, tutorials, and academic references that support the tools and techniques employed during the internship.

2.1 Power BI Overview

Power BI, developed by Microsoft, is a widely used business intelligence tool that enables users to connect, model, and visualize data with ease. As per Microsoft's official documentation [1], Power BI allows seamless integration with various data sources and supports real-time dashboarding, making it ideal for fast-paced business environments.

In their book, *Introducing Microsoft Power BI*, Ferrari and Russo [2] explain the core concepts of Power BI, including DAX formulas, data modeling, and dashboard design, offering a strong foundation for building effective reports.

Online resources like JavaTpoint's Power BI tutorial [3] provide beginner-to-advanced-level training with step-by-step guidance on data loading, transformation, and visual report building. Moreover, NovyPro's showcase dashboards [4], such as EV Sales and Retail Analytics, demonstrate practical applications of Power BI across sectors.

These references closely align with the dashboards developed during my internship in GDP comparison, EV sales trend visualization, and e-commerce sales analysis.

2.2 Python for Data Cleaning and Analysis

Python is a versatile programming language heavily used in data science for its powerful libraries and simplicity. Cao (2017) [5], in *ACM Computing Surveys*, outlines how Python's ecosystem— particularly libraries like Pandas, NumPy, and Matplotlib—enables robust data manipulation and visualization.

For beginners, W3Schools and Geeks for Geeks offer practical tutorials on handling datasets with Python, from removing null values to grouping and summarizing large records [6][7]. These were

particularly useful during preprocessing stages in the GDP and EV datasets.

Raschka et al. (2020) [8], in their comprehensive survey, also highlighted the growing role ofPython in both exploratory data analysis and pipeline integration with tools like Power BI.

2.3 Literature Supporting Domain-Specific Applications

GDP Analysis

Analyzing economic indicators like GDP requires both statistical understanding and clear presentation. A study by Yuan et al. (2019) [9], published in the *Journal of Economic Structures*, emphasizes the role of data visualization in making cross-country GDP trends easier to understand. They recommend using visuals like bar and line charts to communicate long-term growth patterns—a practice mirrored in my internship dashboard.

Electric Vehicle (EV) Sales

EV adoption analytics is gaining prominence due to its relevance in climate action and mobility planning. Lutsey and Nicholas (2019) [10], in an ICCT report, discuss EV growth and policy across India and Asia. Their use of state-level comparisons and growth metrics helped guide the structure of my EV dashboard.

E-Commerce Sales Dashboarding

E-commerce analytics helps track product performance and customer preferences. In the *Journal of Business Research*, Wamba et al. (2017) [12] confirmed that data visualization in retail leads to better customer targeting and sales growth.

These studies align with the logic and layout applied in my e-commerce dashboard, where dynamic filters and DAX-based KPIs were used to analyze top-performing products and regions.

CHAPTER 3

METHODOLOGY

Power BI is a business intelligence tool that helps users visualize data through interactive dashboards and reports. It simplifies data analysis and enables organizations to monitor key metrics in real-time.

3.1 Technologies Learned

Python is a versatile language used for data manipulation, visualization, and machine learning. I worked with key libraries such as:

- NumPy & Pandas: For numerical operations and data manipulation.
- Matplotlib & Seaborn: For visualizing data and identifying trends.
- Data Cleaning: Handled missing values, outliers, and formatting issues.
- Exploratory Data Analysis (EDA): Explored datasets to find patterns and anomalies.
- Statistical & Predictive Analysis: Applied statistical methods and basic machine learning models for insights and forecasting.

3.2 Tasks Assigned During Training Program

The methodology highlights the systematic approach to data collection, preprocessing, analysis, and visualization across various projects. These efforts culminated in the development of interactive dashboards and comprehensive reports, providing actionable insights for datadriven decision-making.

1. Madhav Ecommerce Sales Dashboard

During my internship, I developed an interactive dashboard for Madhav, an e-commerce platform, to track and analyze online sales data. The primary goals were:

- Monitor Sales Performance: Across different categories and periods.
- Analyze Customer Purchasing Patterns: To understand customer behavior.
- Identify Top-Performing Products and Sales Trends: For strategic decisions.
- Make Data-Driven Decisions: To improve marketing and sales strategies.

Implementation

- **Bar Charts:**
    - Sum of Profit by Sub-Category – Displays profitability across product lines such as

Printers, Bookcases, and Sarees.

- Sum of Amount by State – Visualizes state-wise revenue contributions (e.g., Maharashtra, MP).

- **Stacked Column Charts:**
  - Profit by Month – Shows monthly profit trends, highlighting seasonality.
  - Sum of Amount by Customer Name – Displays top-spending customers like Hainvansh and Madhav.
- **Donut Charts:**
  - Sum of Quantity by Category – Represents quantity distribution across Clothing, Electronics, and Furniture.
  - Sum of Quantity by Payment Mode – Highlights preferred payment methods like COD and Debit Card.
- **Card Visuals:**
  - Displays key performance indicators:
    - Sum of Amount (₹438K)
    - Sum of Quantity (6K)
    - Sum of Profit (₹37K)
- **Slicers:**
  - Quarter Filter – Enables users to filter data by Q1, Q2, Q3, or Q4.
  - State Filter – Allows selection of individual states for focused analysis.

EV Sales Analysis Dashboard:

During my internship I have build EV Sales Analysis Dashboard.The primary goals are:

- **Analyze EV Adoption Trends**:
  Track and visualize the growth of electric vehicle (EV) adoption across years, quarters, and months—categorized by state, manufacturer, and vehicle type (2W/4W).
- **Identify Top Performers and Opportunities:**
  Determine the top-selling EV makers, states with high and low penetration rates, and growth patterns using KPIs and comparison visuals to inform strategic decisions.
- **Enable Interactive Exploration of Data:**
  Build an interactive Power BI dashboard with filters and slicers to allow dynamic exploration of data by users, supporting data-driven policy and market strategies.

Implementation Details

- **Card Visuals:**
  - Key metrics such as CAGR (93.91%), Total EVs Sold (2M), Revenue by 2W and 4W Makers (176bn & 3T), and Penetration Rate (3.61) were highlighted using KPI cards for quick, high-level insight.

- **Map Visual:**
  - Sum of EVs Sold by State represented with a Map chart to give a spatial distribution of EV sales across India.
- **Line Charts:**
  - EVs Sold by Quarter and Maker – Trend analysis of key manufacturers (BYD India, Hyundai, Tata, etc.) across different quarters.
  - EV Sales by Month – Monthly sales patterns across different months of the year.
- **Column Charts:**
  - EVs Sold by Year – Year-wise breakdown from 2021 to 2024.
  - EVs Sold by Fiscal Year and Maker – Trends in total vehicles sold by top companies across fiscal years.
- **Donut Charts:**
  - EVs Sold by Vehicle Category – Breakdown between 2-Wheelers and 4-Wheelers.
  - Delhi vs Karnataka Penetration Rate (2024) – Comparison of regional adoption rates using a double donut.
- **Bar Charts:**
  - Top 3 Makers vs Bottom 3 Makers – Identifying leading and lagging companies based on units sold.
  - Top 5 States by 2-Wheeler and 4-Wheeler Penetration Rate (FY 2024*)* – Highlighted adoption hotspots for both categories.
- **Slicers:**
  - Enabled interactive filtering by Year (2021–2024) and Vehicle Category (2W / 4W) to allow users to explore data dynamically.

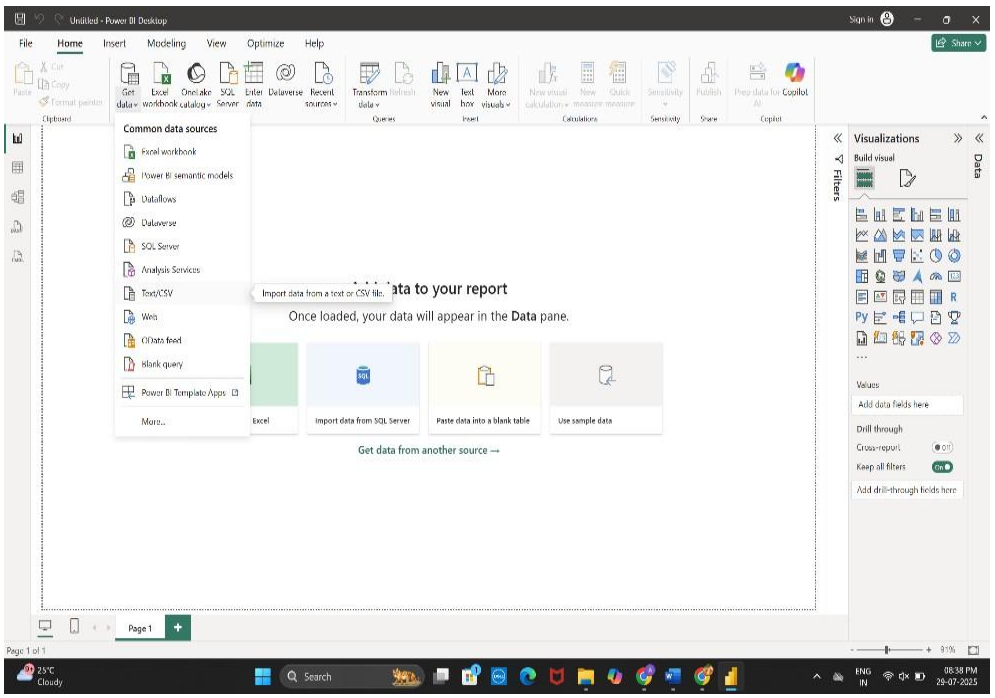
3.3 **Process for the Analysis**

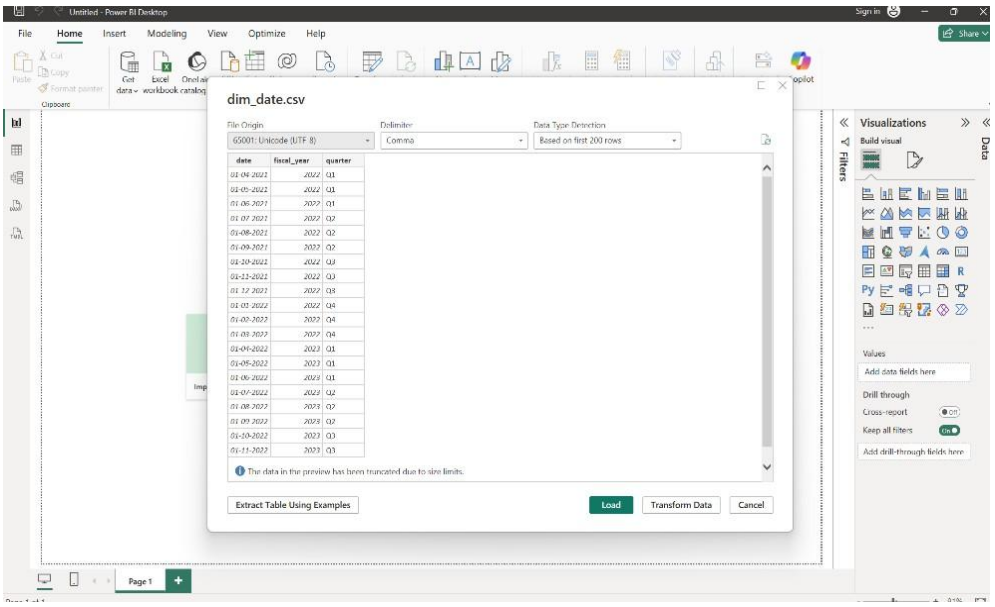
First, you need to install a Power BI desktop on your PC or laptop. The link to install the Power BI Desktop is provided below:-

https://youtu.be/GT2NcTE6UEo?feature=shared

Import the Data

After installing Power BI, we need to import the downloaded file into the tool. By clicking the option of get data you can import files into the tool. For this analysis, we will go with the Text/CSV option for uploading the file.

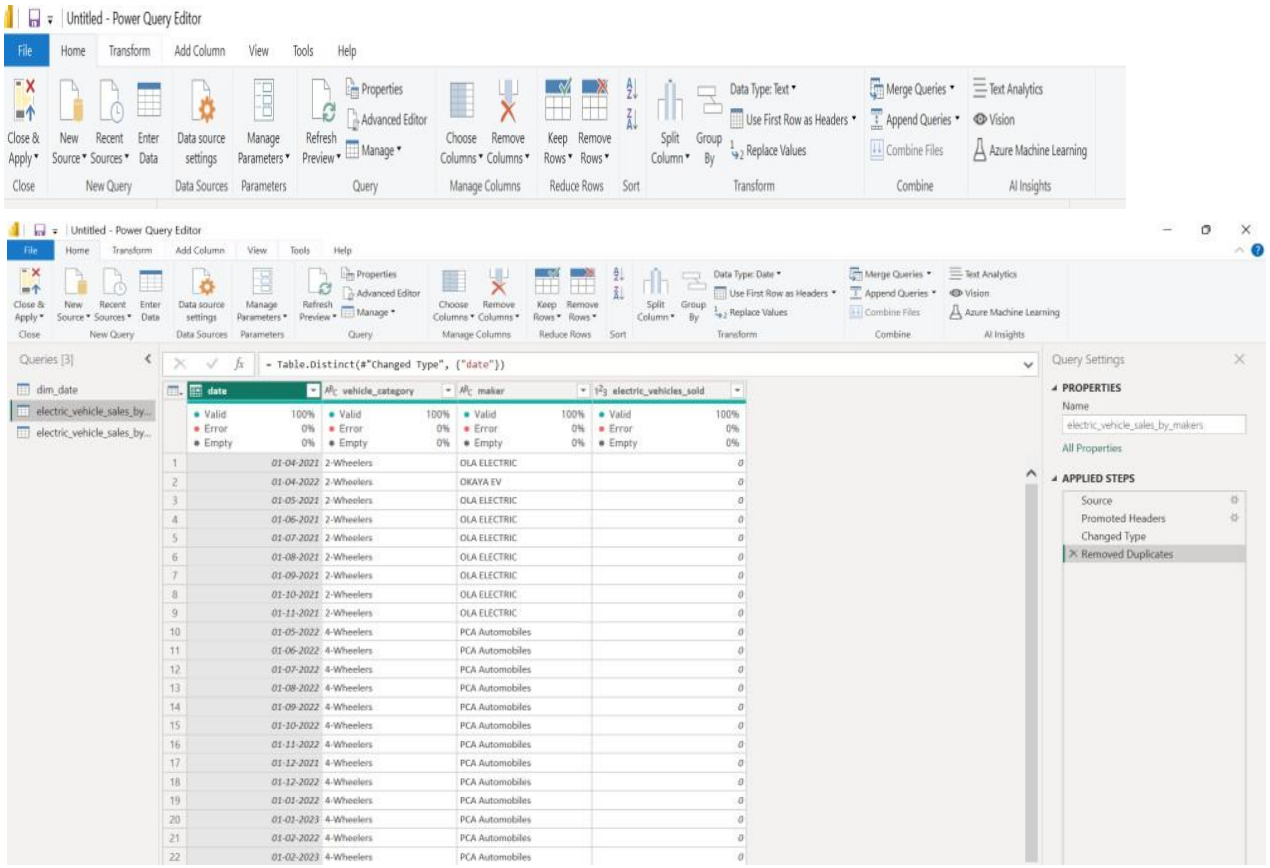




Clean the data

After loading a file into the tool, let's clean the data which we have imported. Power Query editor is a very useful tool in power bi, it helps to clean the data. To go into the power query editor click on the option of Transform data. Click on the Remove rows option and remove the duplicates from the table. This process will remove all the duplicate values present in the table . We can make a lot more changes in the data with the help of a power query editor. To go back to the dashboard page click on the close

& apply icon. It will close the power query editor by saving the changes and get back to you on your dashboard page of the tool .

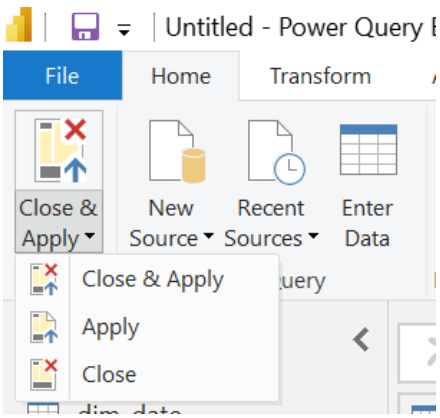


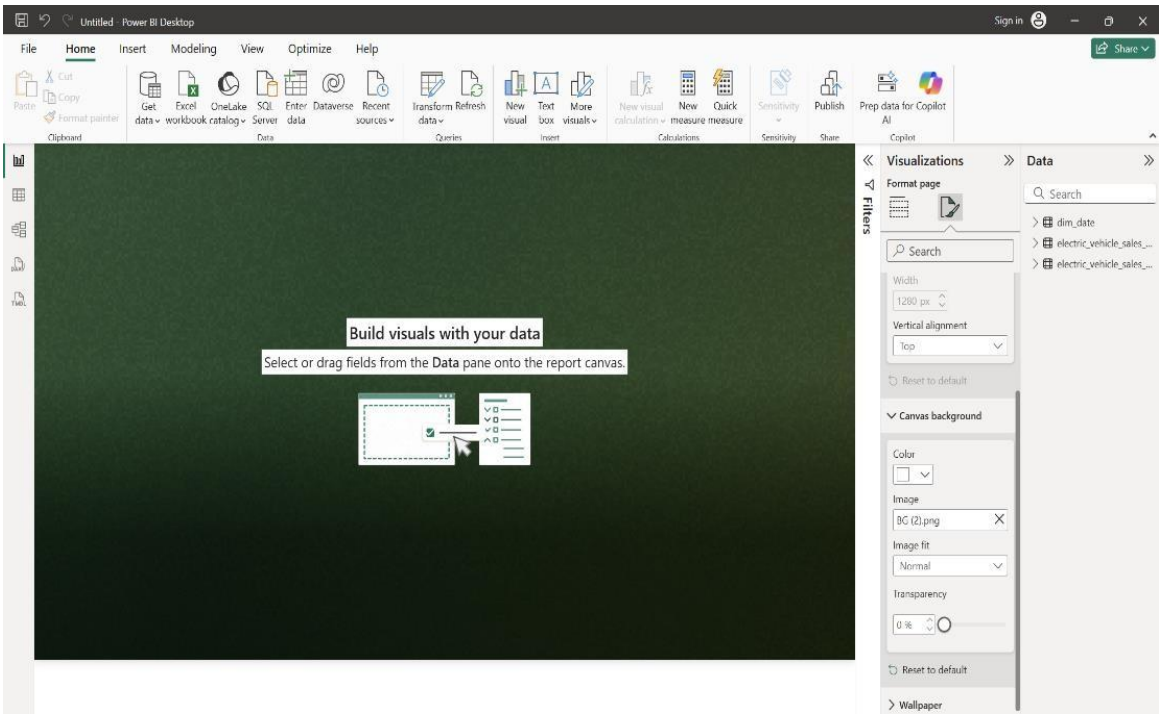
Use the data

To change the background of the dashboard click on the blank space in the dashboard then go to the format option which is under the format page, then look for the wallpaper option and browse the background of your choice .

Once the image is added, adjust the Transparency setting to 0% to make the image fully opaque or 100% to make it transparent.

Once the data is loaded in the tool, you will be ready to create visuals from the data.



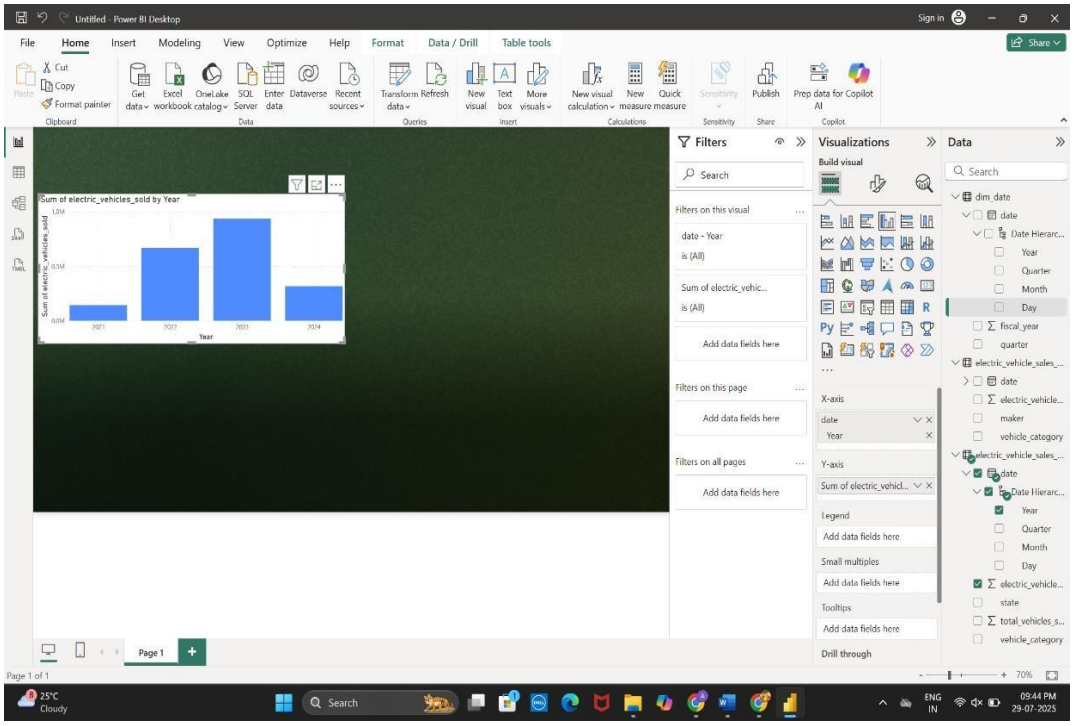


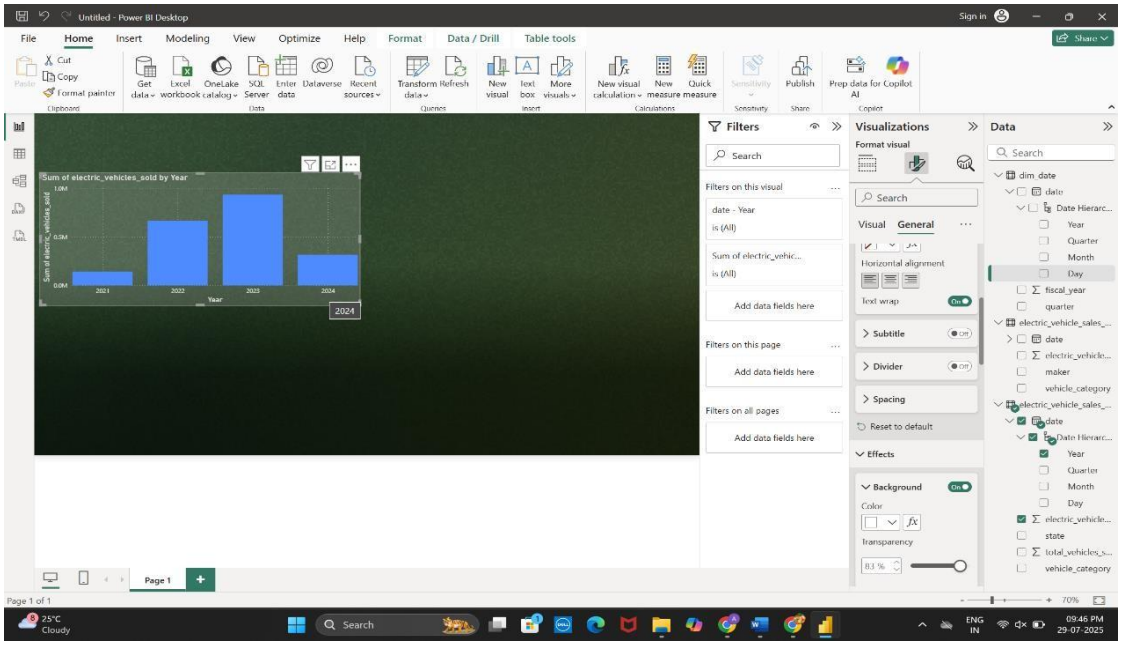
Title to Dashboard

To add a title to the dashboard go to the insert tab and select the Text box icon.

It will paste a box on the dashboard area in which you can give a title to your dashboard .

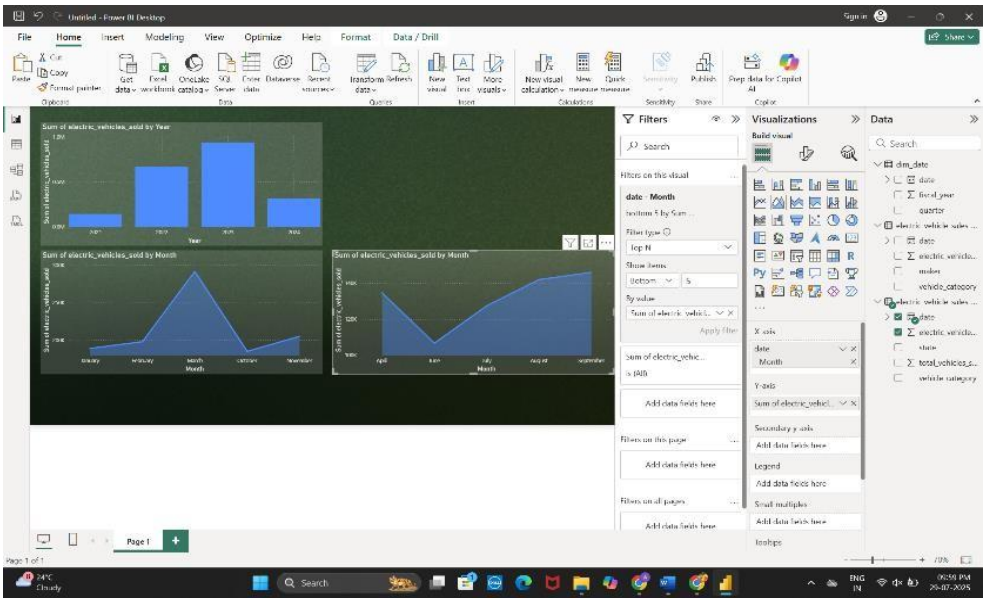
Changing the background of a visual or adjusting the transparency of a background is useful for blending visuals with the report page's background





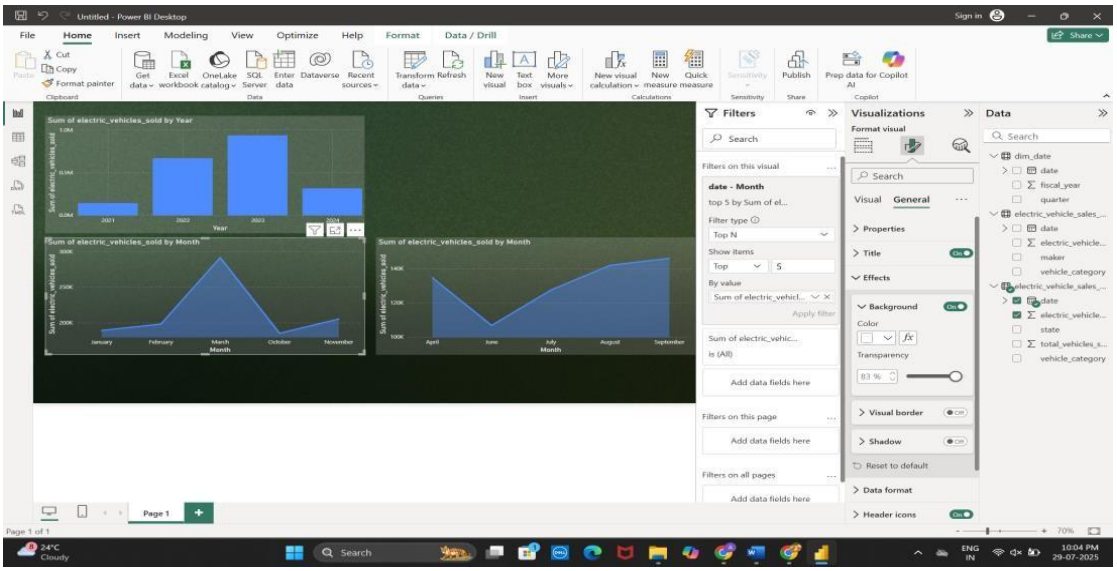
Filters

To make this visual more understandable we will use the filter on this visual. To use the filter drag the sum of electric vehicles sold in the filters section select the Top N filter and put the values of date-Month.



Edit the graph

To edit the graph go to the format option of visual and change the title, background transparency, font color, To make the graph background transparent and to make the rounded border, you need to go to the format section of the visual and select general, there you will find the option of effect in which you can see the background and visual border.



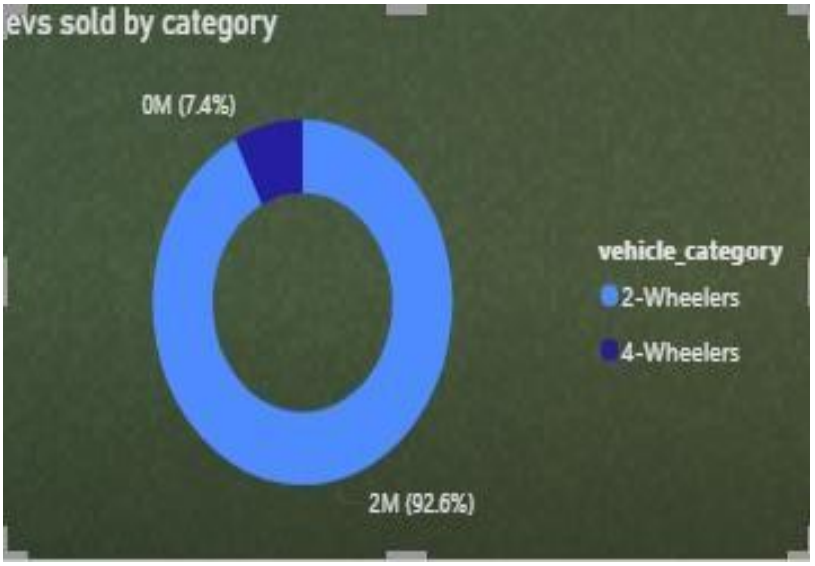
Analysis & Explanation

- CAGR (93.91%): The Compound Annual Growth Rate shows rapid EV market expansion.
- Total EV Sold (2M): Two million electric vehicles were sold over the analyzed period.
- Revenue 2W Makers (176bn): Two-wheeler EV manufacturers generated ₹176 billion in revenue.
- Revenue 4W Makers (3T): Four-wheeler EV manufacturers achieved ₹3 trillion in revenue.
- Penetration Rate (3.61): EVs represent 3.61% of total vehicle sales, indicating early-stageadoption.
- Year Slicer (2021–2024): Lets users filter dashboard visuals by specific years.

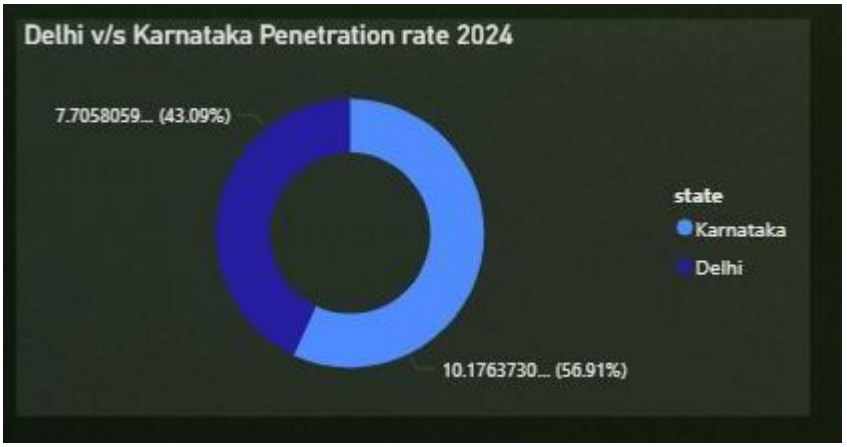
- Tata Motors had the highest sales, though they declined steadily from Q4 to Q1.

- Mahindra & Mahindra saw a slight increase in Q1 after consistent sales.
- Other makers like MG Motor, Hyundai, and BYD India maintained lower, steady volumes.

  Insight: Tata's sales dominate but are declining; Mahindra shows a potential growth trend.

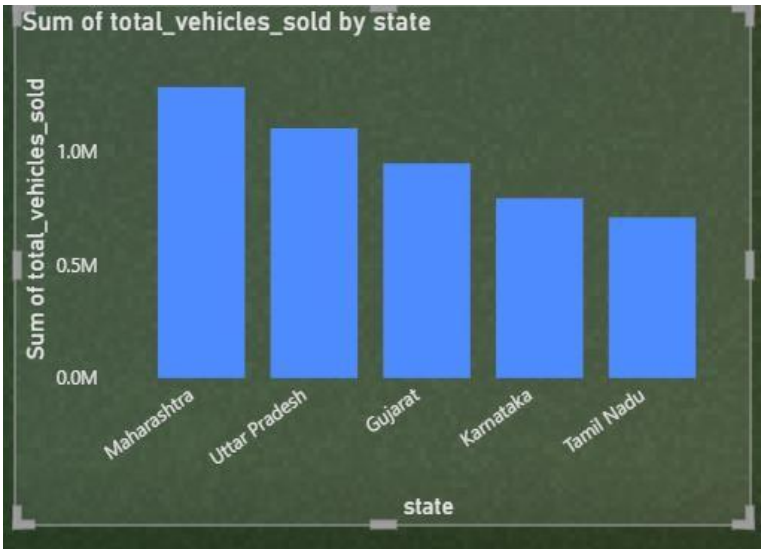


- 2-Wheelers dominate the market, accounting for 92.6% of total EV sales (2 million units).
- 4-Wheelers make up only 7.4% of total sales.



- Karnataka has a higher penetration rate at 10.17 (56.91%).
- Delhi follows with 7.71 (43.09%).

  Insight: Karnataka leads in EV adoption compared to Delhi in FY 2024.



- Maharashtra leads with the highest vehicle sales, followed by Uttar Pradesh and Gujarat.
- Karnataka and Tamil Nadu have relatively lower sales among the top five.

  Insight: Maharashtra and Uttar Pradesh are major markets for vehicle sales in India.

Use Case of this Dashboard

Dashboards make complex data easier to understand through visuals and graphs, helping in quick decision-making.

- E-Commerce Sales Dashboard:

  Useful for retail businesses to track sales, customer behavior, and product performance. It helps improve marketing, inventory, and customer targeting strategies.
- Electric Vehicle Sales Dashboard:

  Beneficial for government bodies and EV manufacturers to monitor adoption trends, penetration rates, and regional performance.

3.4 GDP Analysis Using Python

Problem Statement:

The goal is to study how India's economy is growing by looking at what makes up the GDP, how different sectors contribute, and how things like inflation, government spending, foreign trade, and population growth affect the economy. This helps us understand what drives growth and where improvements can be made.

We explored diverse macroeconomic questions related to:

Economic Structure: Sector-wise contributions, export-import roles, energy usage

Policy Impact: Taxation, fiscal deficit, repo rates, government role

Socioeconomic Trends: Unemployment, inflation, population growth

Pandemic Effects: Pre vs post-COVID GDP shifts

Investment & Growth: FDI inflows and GDP relationship

Dataset information

```
]: df.head()
```

]:

	Year	GDP (Nominal)	GDP (Real)	GDP growth (annual %)	Agriculture Contribution (% of GDP)	Industry Contribution (% of GDP)	Services Contribution (% of GDP)	Inflation CPI	Unemployment Rate (%)	Exports (% of GDP)	...	Population Growth Rate (annual %)	Oil rents (% of GDP)	Government Revenues (% of GDP)	Househ Consumpti Expendit (% of GD
0	1971	35.58	67.4	1.642930	34.58	26.47	43.32	5.324841	5.3	3.667205	...	5.307963	0.087218	9.74	5
1	1972	36.06	71.5	-0.553301	34.44	26.48	43.74	10.839600	5.3	4.027489	...	5.678821	0.078421	9.63	5
2	1973	38.32	85.5	3.295521	34.06	25.77	44.11	17.829720	5.0	4.208763	...	5.705386	0.077427	9.58	5
3	1974	38.47	99.5	1.185336	34.23	25.85	43.77	16.667520	5.2	4.831321	...	5.712841	0.426931	9.90	6
4	1975	41.57	98.5	9.149912	35.26	25.19	43.52	-1.648680	4.9	5.647062	...	5.655836	0.534883	10.09	6

5 rows × 26 columns

We have imported the dataset and checked the shape of the dataset .It has 53 rows and 26 columns.

Data Cleaning

Data Cleaning is a very important step in data analysis, because if we have missing values, duplicate values, or anything like that in our dataset then we will not get the desired output or the information we want so the first & most important step is to clean our data to get meaningful insights, so here we will start cleaning our data.

```
print("Missing values before removal:")
print(df.isnull().sum())
df.dropna(inplace=True)
print("Missing values after removal:")
print(df.isnull().sum())
```

```
Missing values before removal:
Year                        0
GDP_Nominal                 0
GDP_Real                    0
GDP_Growth                  0
Agri_GDP%                   0
Industry_GDP%               0
Services_GDP%               0
Inflation_CPI               0
Unemployment_Rate           0
Exports_GDP%                0
Imports_GDP%                0
FDI_Inflows                 0
Fiscal_Deficit              0
Public_Debt                 0
Repo_Rate                   0
Forex_Reserves              0
Population_Growth           0
Oil_Rents                   2
Govt_Revenue                0
Household_Consumption       0
Capital_Formation           0
Private_Credit              0
Poverty_Rate                0
Energy_Consumption          0
Current_Account_Balance     0
Tax_Revenue                 0
dtype: int64
```

Identified 2 missing values found in Oil_Rents using df.isnull().sum().so we will drop them, because we have 53 rows we can simply drop those 2 rows.

```
Missing values after removal:
Year                        0
GDP_Nominal                 0
GDP_Real                    0
GDP_Growth                  0
Agri_GDP%                   0
Industry_GDP%               0
Services_GDP%               0
Inflation_CPI               0
Unemployment_Rate           0
Exports_GDP%                0
Imports_GDP%                0
FDI_Inflows                 0
Fiscal_Deficit              0
Public_Debt                 0
Repo_Rate                   0
Forex_Reserves              0
Population_Growth           0
Oil_Rents                   0
Govt_Revenue                0
Household_Consumption       0
Capital_Formation           0
Private_Credit              0
Poverty_Rate                0
Energy_Consumption          0
Current_Account_Balance     0
Tax_Revenue                 0
dtype: int64
```

Identify duplicate values and removing them

```
print("Duplicate rows before removal:", df.duplicated().sum())
df.drop_duplicates(inplace=True)
print("Duplicate rows after removal:", df.duplicated().sum())
```

```
Duplicate rows before removal: 0
Duplicate rows after removal: 0
```

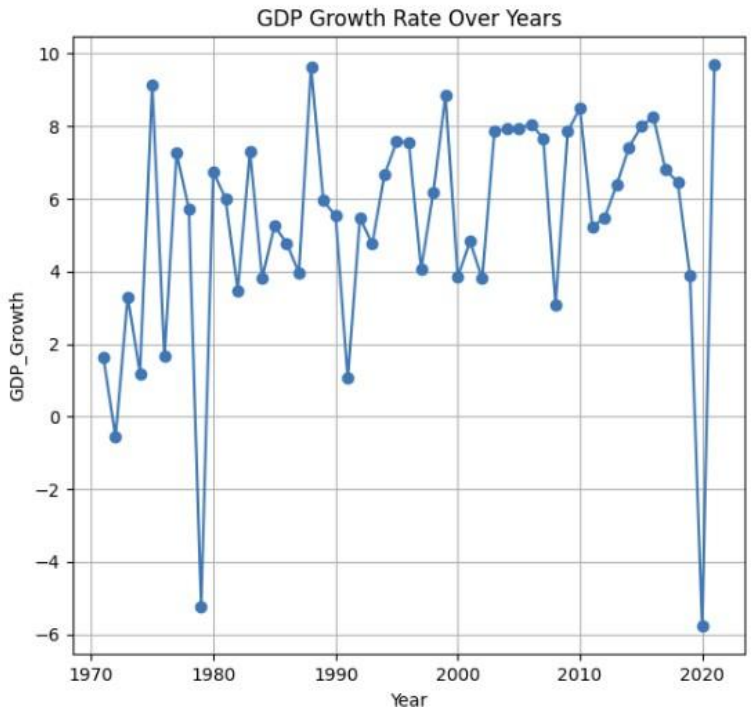
```python
# Rename long columns to short, typo-safe names
df.rename(columns={
    'GDP (Nominal)': 'GDP_Nominal',
    'GDP (Real)': 'GDP_Real',
    'GDP growth (annual %)': 'GDP_Growth',
    'Agriculture Contribution (% of GDP)': 'Agri_GDP%',
    'Industry Contribution (% of GDP)': 'Industry_GDP%',
    'Services Contribution (% of GDP)': 'Services_GDP%',
    'Inflation CPI': 'Inflation_CPI',
    'Unemployment Rate (%)': 'Unemployment_Rate',
    'Exports (% of GDP)': 'Exports_GDP%',
    'Imports  (% of GDP)': 'Imports_GDP%',
    'FDI Inflows (USD)': 'FDI_Inflows',
    'Fiscal Deficit (% of GDP)': 'Fiscal_Deficit',
    'Public Debt (% of GDP)': 'Public_Debt',
    'Repo Rate (%)': 'Repo_Rate',
    'Foreign Exchange Reserves (MILLION USD)': 'Forex_Reserves',
    'Population Growth Rate (annual %)': 'Population_Growth',
    'Oil rents (% of GDP)': 'Oil_Rents',
    'Government Revenues (% of GDP)': 'Govt_Revenue',
    'Household Consumption Expenditure (% of GDP)': 'Household_Consumption',
    'Gross Fixed Capital Formation (% of GDP)': 'Capital_Formation',
    'Private Sector Credit (% of GDP)': 'Private_Credit',
    'Poverty Rate (%)': 'Poverty_Rate',
    'Energy Consumption per Capita (kWh)': 'Energy_Consumption',
    'Current Account Balance (% of GDP)': 'Current_Account_Balance',
    'Tax revenue (% of GDP)': 'Tax_Revenue'
}, inplace=True)

print(df.columns.tolist())
```

```
['Year', 'GDP_Nominal', 'GDP_Real', 'GDP_Growth', 'Agri_GDP%', 'Industry_GDP%', 'Services_GDP%', 'Inflation_CPI', 'Unemployment_Rate', 'Exports_GDP%', 'I
mports_GDP%', 'FDI_Inflows', 'Fiscal_Deficit', 'Public_Debt', 'Repo_Rate', 'Forex_Reserves', 'Population_Growth', 'Oil_Rents', 'Govt_Revenue', 'Household
_Consumption', 'Capital_Formation', 'Private_Credit', 'Poverty_Rate', 'Energy_Consumption', 'Current_Account_Balance', 'Tax_Revenue']
```

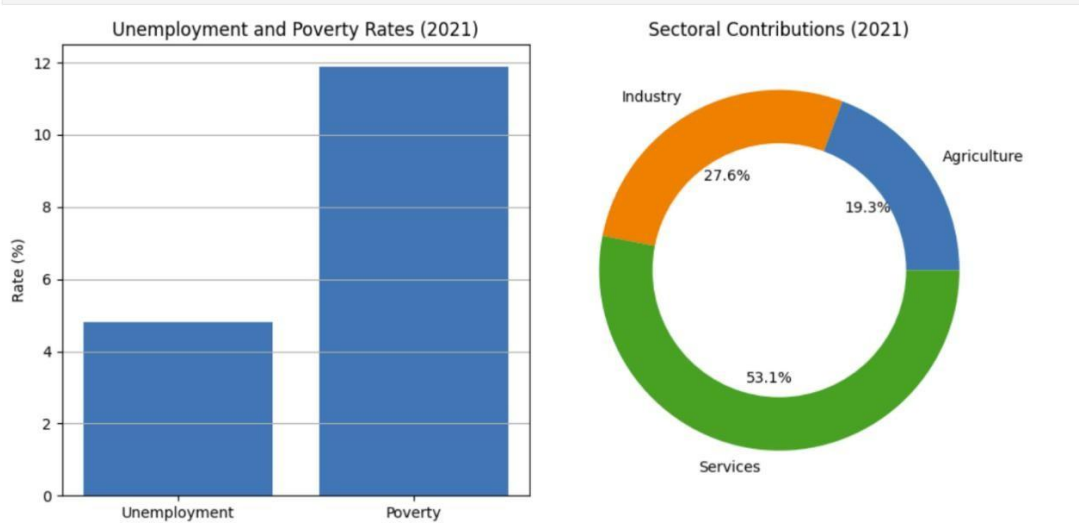
No duplicates were found and we have renamed the columns for simplicity.

Exploratory Data Analysis:

Univariate analysis

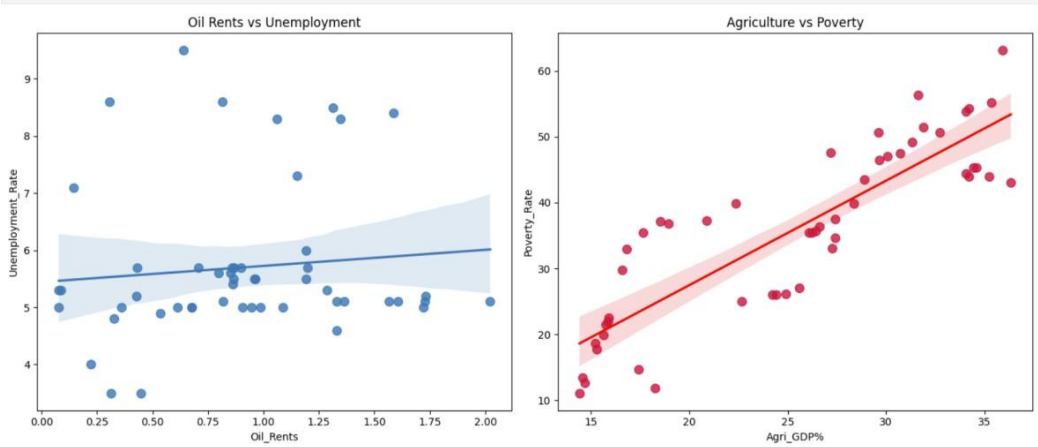


The graph shows that India's GDP growth has mostly been positive with some sharp declines during crises like in 1980, 1991, and 2020.

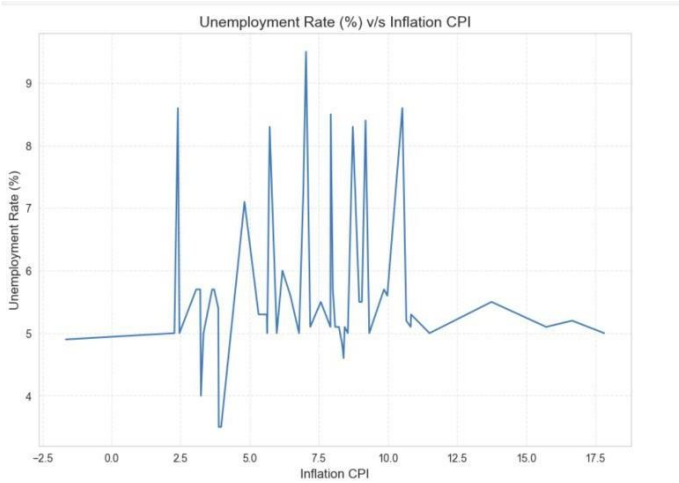


- Unemployment rate is much lower than poverty rate signifying earnings are not sufficient to come out of poverty line
- Service sector dominates towards contribution which is 53.1% and Agriculture sector contribution is least 19% showing service-driven economy.

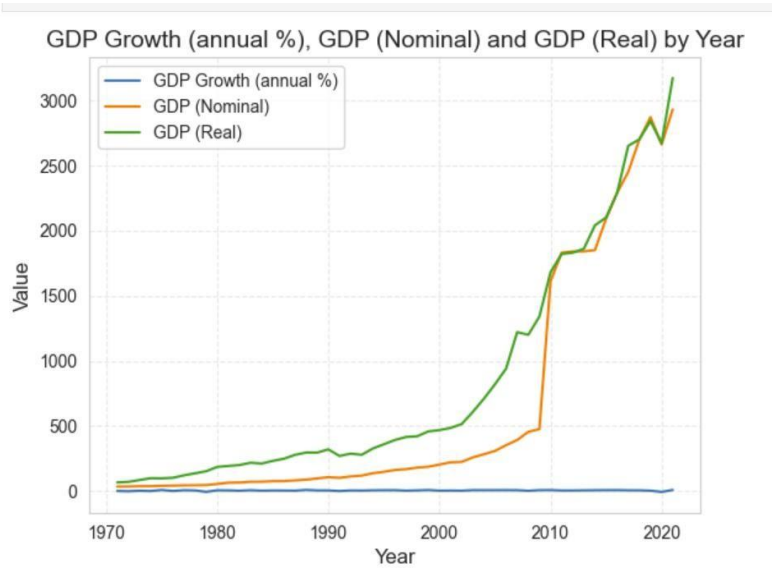
Bivariate Analysis



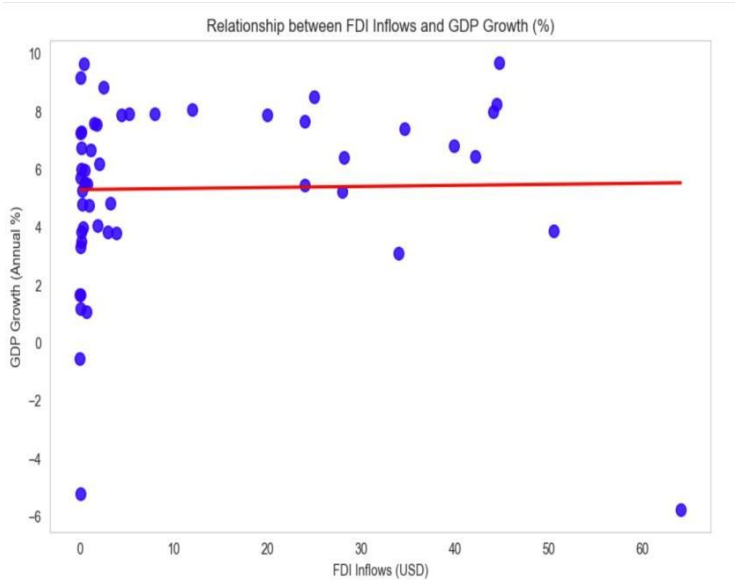
The first graph shows little to no clear relationship between oil rents and unemployment, while the second graph shows that higher agricultural GDP is associated with higher poverty rates.



The graph shows no clear or consistent relationship between inflation (CPI) and unemployment rate.



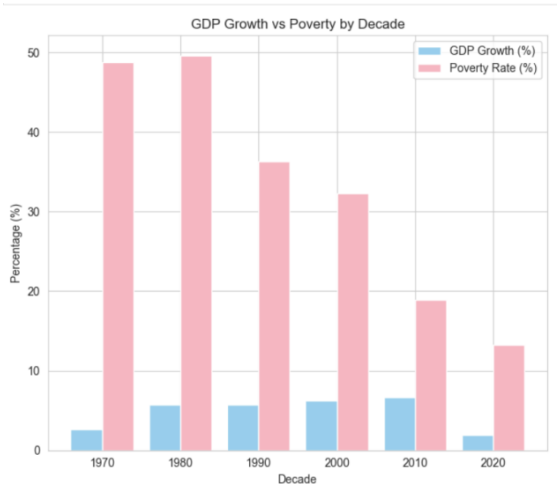
The graph shows that while GDP Growth (%) remained fairly steady, both Nominal and Real GDP have increased sharply over the years, especially after 2005.



The graph shows a very weak or no clear relationship between FDI inflows and GDP growth rate.

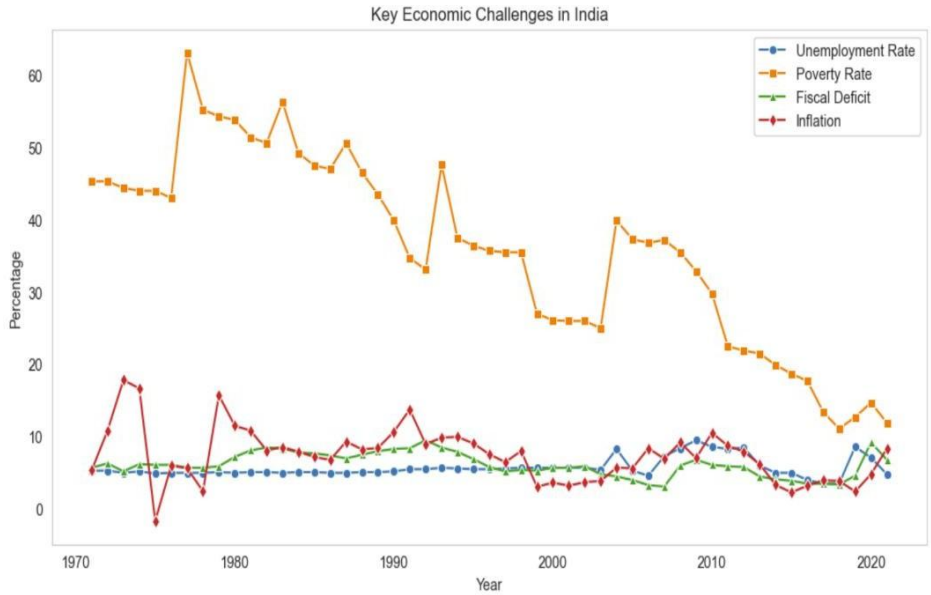
The graph shows an overall increasing trend in both exports and imports' contribution to India's GDP from 1970 to around 2010, followed by a decline towards 2020.

Multivariate analysis

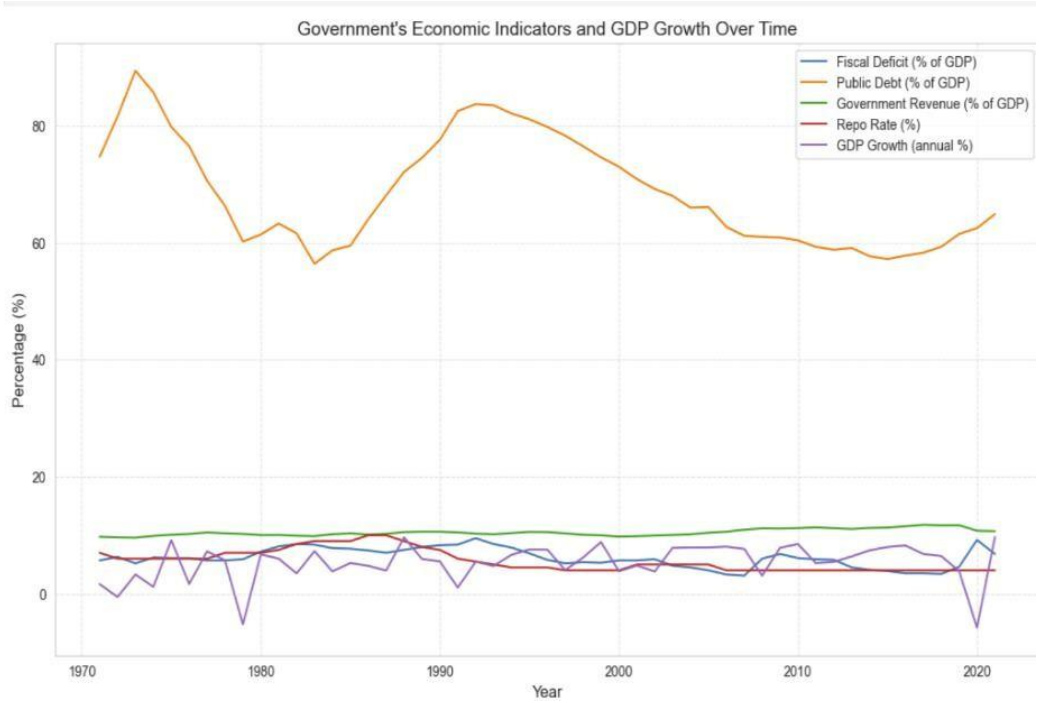


The graph shows a general decline in poverty rate alongside a fluctuating but generally lower GDP

growth rate across decades from 1970 to 2020.



The graph shows a general decline in poverty rate and fiscal deficit, a fluctuating unemployment rate with a slight increase towards 2020, and a variable inflation rate with peaks in the 1970s and 2010s over the period from 1970 to 2020.



The graph shows a high and fluctuating fiscal deficit peaking in the 1980s, a relatively stable public debt and government revenue around 20-30% of GDP, and a volatile GDP growth with a notable decline around 2020.

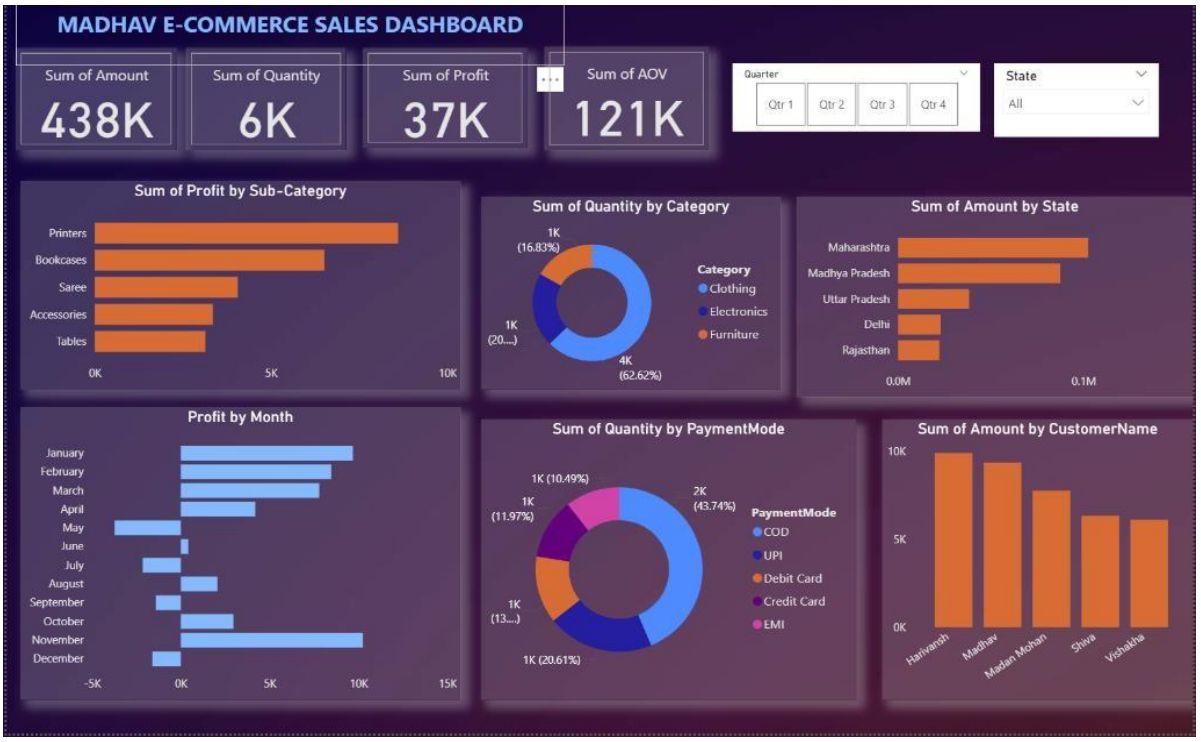
CHAPTER 4

RESULT AND DISCUSSION

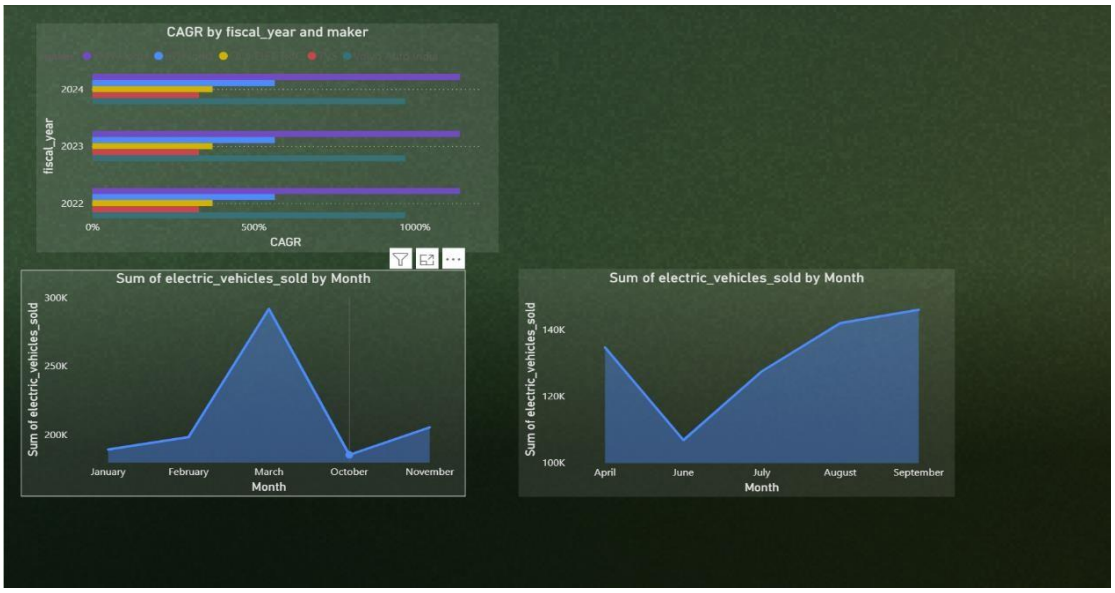
Throughout my internship at the Indian Institute of Information Technology, Nagpur (IIIT Nagpur), I achieved significant results by leveraging data analytics and visualization tools to derive actionable insights across various domains. Utilizing Microsoft Power BI, I developed dynamic and interactive dashboards that provided deep insights into e-commerce trends, financial analysis, mobile app performance, restaurant assessments, video game sales, and supply chain management. My Python projects, including Diwali Sales Analysis and Student Exam Scores Analysis, further demonstrated my ability to analyze complex datasets, identify key trends, and provide strategic recommendations. These results underscored my proficiency in data pre-processing, statistical analysis, and visualization, ultimately supporting data-driven decision-making and optimizing business strategies.

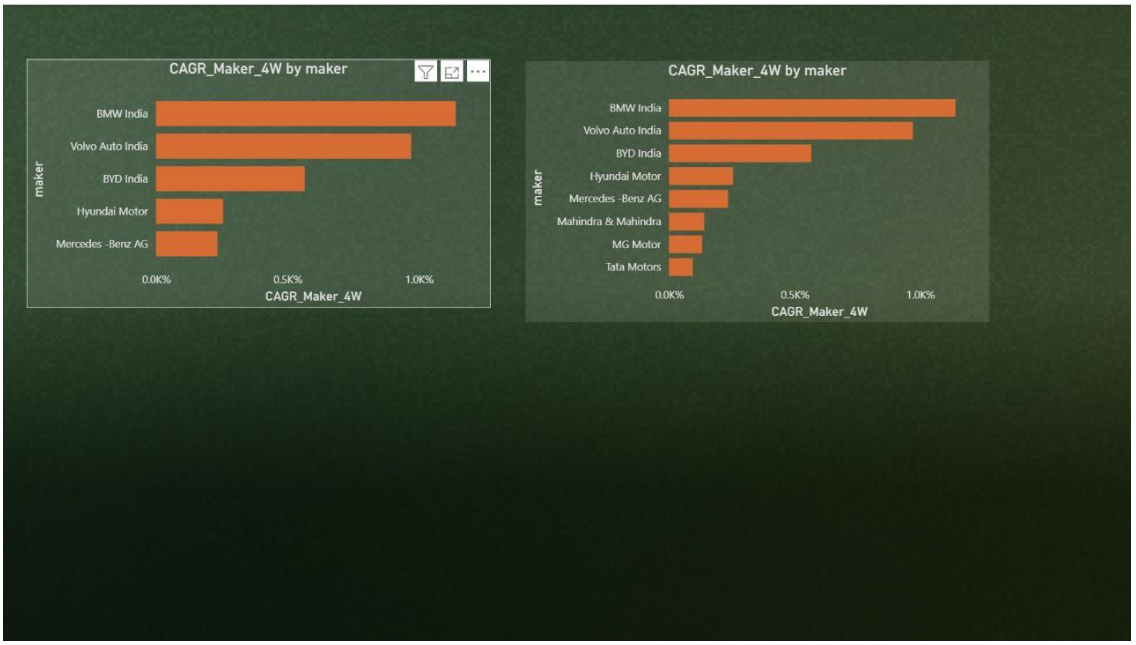
4.1 Outcome of Data Analysis

1 Madhav E commerce sales dashboard



2 EV Sales Analysis Dashboard





4.2 Internship Experience Summary

During my internship at the Indian Institute of Information Technology, Nagpur (IIIT Nagpur), I worked on real-world data analytics projects across domains like e-commerce and electric vehicle sales. Using tools like Power BI and Python, I performed data cleaning, analysis, and dashboard creation to derive meaningful insights.

This experience helped me apply classroom concepts to practical scenarios, strengthen my problem-solving abilities, and improve my skills in data visualization and decision-making. Each project brought new challenges and learnings, making this internship a valuable step in my data analytics journey.

4.3 How the Objectives Were Achieved

The goal of the internship was to turn raw data into clear, useful insights—and I achieved that through hands-on work with Python and Power BI. For the EV Sales project, I cleaned and prepared data, then built a dashboard that showed which states and makers were leading the market.

In the E-commerce project, I used Power BI to track sales, customer behavior, and product performance. Using features like slicers, KPIs, and DAX, I made the data easy to explore and understand. Every project helped me apply what I've learned in a practical, meaningful way.

4.4 Skills Learned

Throughout the internship, I developed a strong combination of technical and professional skills:

- Technical Skills:

  I became proficient in using Power BI for data connection, transformation, and visualization. I also learned to write DAX formulas for custom metrics and use Power Query Editor for data preparation. In Python, I gained experience with data manipulation using pandas, as well as visualization using matplotlib and seaborn.
- Analytical and Statistical Thinking:

  Each project required a structured analytical approach, helping me strengthen my ability to work with large datasets, identify patterns, and draw meaningful insights.
- Professional Skills:

  The internship also enhanced my ability to manage timelines, communicate results effectively, and create user-friendly dashboards. These skills were essential in simulating real-world business and policy analysis tasks.

4.5 Results, Observations, and Work Experiences

Each project resulted in valuable findings that showcased the impact of thoughtful data analysis:

- In the EV Sales Analysis, I observed that states like Maharashtra and Uttar Pradesh led in overall vehicle adoption, and penetration rates were increasing consistently year over year. These insights can guide manufacturers and policymakers in resource allocation and infrastructure planning.
- The E-Commerce Dashboard revealed top-selling products, customer behavior by region, and seasonal sales trends. Such insights are critical for inventory planning, marketing, and customer

retention strategies.

- The GDP Analysis uncovered sectoral contributions to national GDP and highlighted growth patterns. This offered a macroeconomic perspective useful for analysts and researchers studying economic performance.

4.6 Challenges Experienced

The internship posed a number of practical challenges that helped me grow technically and professionally:

- In the EV Sales project, managing diverse state-wise data and ensuring consistent formats across quarters was challenging. It required careful transformation and validation within Power BI.
- In the E-Commerce Dashboard, creating accurate DAX formulas for custom KPIs and integrating multiple data tables tested my data modeling skills.
- The GDP Analysis involved handling null values, inconsistencies in column formats, and converting large CSV datasets into meaningful time-series visualizations—all using Python. Overcoming these challenges helped me build confidence in handling real-world data scenarios and taught me to think critically and solve problems independently.

CHAPTER 5

CONCLUSION

I gained extensive hands-on experience in data analysis and visualization through a series of practical, real-world projects. My work on the Electric Vehicle (EV) Sales Analysis project involved leveraging Microsoft Power BI to explore market trends, penetration rates, and manufacturer performance. This project enhanced my ability to create dynamic dashboards that offered valuable insights for stakeholders such as EV companies, policymakers, and sustainability advocates.

Similarly, my involvement in the E-Commerce Sales Dashboard allowed me to track customer behavior, product performance, and payment trends using interactive visuals. Through Power BI tools like Power Query and DAX, I was able to transform raw data into actionable insights that could help online retailers make more informed decisions about inventory management and targeted marketing.

Additionally, my GDP Analysis using Python introduced me to economic data interpretation. By applying data preprocessing and exploratory data analysis techniques, I was able to identify patterns in GDP growth, understand contributing factors, and create visualizations that conveyed complex trends in a simplified manner. This project not only broadened my understanding of economic indicators but also strengthened my skills in using Python libraries like pandas, matplotlib, and seaborn.

Each of these projects presented unique challenges and learning opportunities. They allowed me to apply theoretical knowledge to real-world scenarios and sharpened my skills in data cleaning, visualization, interpretation, and storytelling. More importantly, they deepened my understanding of how data-driven insights can drive smarter, more strategic decisions across different industries.

REFERENCES

1. **Microsoft. *Power BI Documentation*. Retrieved from: https://learn.microsoft.com/en-us/power-bi**
2. **Ferrari, A., & Russo, M. (2016). *Introducing Microsoft Power BI*. Microsoft Press.**
3. **JavaTpoint. *Power BI Tutorial*. Retrieved fromhttps://www.tutorialspoint.com/power_bi/index.htm**
4. **NovyPro. *Power BI Dashboard Examples*. Retrieved from: https://www.novypro.com/project**
5. **Cao, L. (2017). Data Science: A Comprehensive Overview. *ACM Computing Surveys (CSUR)*, 50(3), 43. DOI: https://doi.org/10.1145/3076253**
6. **W3Schools. *Python Pandas Tutorial*. Retrieved from: https://www.w3schools.com/python/pandas/default.asp**
7. **GeeksforGeeks. *Data Preprocessing with Python*. Retrieved from: https://www.geeksforgeeks.org/data-preprocessing-in-data-mining/**
8. **Raschka, S., Mirjalili, V., & Liu, Y. (2020). *Python Machine Learning* (3rd ed.). Packt Publishing.**
9. **Yuan, C., Wang, X., & Liu, Y. (2019). Visualizing and analyzing GDP trends across countries. *Journal of Economic Structures*, 8(1), 1–15. DOI:**
10. **Lutsey, N., & Nicholas, M. (2019). *Update on Electric Vehicle Costs in India through 2030*. International Council on Clean Transportation (ICCT). Retrieved from: https://theicct.org/wp-content/uploads/2021/12/E2W-cost-2030-India-jul2021.pdf**
11. **Wamba, S. F., Gunasekaran, A., Akter, S., Ren, S. J. F., Dubey, R., & Childe, S. J. (2017). Big data analytics and firm performance: Effects of dynamic capabilities. *Journal of Business Research*, 70, 356–365. DOI: https://doi.org/10.1016/j.jbusres.2016.08.009**